# EFRAIN RIVAS

SENIOR CLOUD ARCHITECT | DEVOPS & INFRASTRUCTURE SYSTEMS ENGINEER

• CLOUD • DEVOPS • K8S • SYSADMIN ARCHITECT

📍 Cary, NC ⚡ resume.ertechlab.com 🌐 linkedin.com/in/efrain-rivas-93455a10 💻 github.com/digdast

## // 01 PROFESSIONAL PROFILE & ARCHITECTURAL FOCUS

Results-driven **Senior Cloud Architect, DevOps Engineer, and Infrastructure Systems Administrator** with 20+ years of hands-on enterprise experience designing, modernizing, and automating mission-critical systems across Microsoft Azure, High-Availability Kubernetes, Sovereign AI/LLM platforms, Active Directory Domain Services, Linux, and multi-hypervisor SAN clusters. Proven architectural track record deploying global multi-site Active Directory forests across 7 data centers, directing 5-stage financial CI/CD release pipelines with 99.999% SLA uptime, orchestrating a 10-node high-availability Kubernetes (K3s) cluster (5-node etcd quorum + 5 workers, 72 vCPUs / 84GB RAM, 70+ pods across 14 namespaces), engineering dual-GPU local AI & multi-agent inference platforms (90+ tok/sec Flash-Attention), and implementing zero-touch endpoint provisioning via Microsoft Intune and Windows Autopilot. Creator of open-source infrastructure & DevOps platforms (**NovaFlix, AegisDNS, SynapseFlow, Pulsearr, Renamarr, PrintForge, Proxmox Cluster Suite, 58+ Intune remediation modules**) with deep expertise in PowerShell 7, Python (Asyncio/FastAPI), Terraform, Ansible, and Zero Trust identity governance.

## // 02 CORE TECHNICAL COMPETENCIES & SYSTEMS MATRIX

### ▶ CLOUD & HYBRID IDENTITY ARCHITECTURE

`Microsoft Azure (IaaS/PaaS)` `Entra ID (Azure AD)`
`Active Directory (7 DCs)` `Hybrid Identity (PHS/PTA)`
`VNets / NSGs / AKS` `Azure Files / AVD / FSLogix`
`Group Policy (GPO) / DFSR` `M365 & Exchange Online`

### ▶ CONTAINERS, GITOPS & HA KUBERNETES

`10-Node K3s / K8s HA Cluster` `5-Node Control-Plane etcd`
`Traefik & NPM Ingress` `Cloudflared DaemonSets`
`70+ Pods / 14 Namespaces` `NVMe HostPath & NFS Storage`
`Klipper ServiceLB` `Sysctl Kernel Tuning`

### ▶ VIRTUALIZATION, HYPERVISORS & SAN

`Proxmox VE (5-Node HA)` `VMware vSphere / ESXi`
`Enterprise SAN (iSCSI/FC)` `Linux KVM & Hyper-V`
`Citrix XenServer / XCP-NG` `MPIO & Failover Clustering`
`AVX2 Host Passthrough` `Zero-Balloon Memory Tuning`

### ▶ DEVOPS, IAC & SOVEREIGN AI PLATFORMS

`PowerShell 7+ & Graph API` `Python (Asyncio/FastAPI)`
`CI/CD Release Pipelines` `Ansible & Terraform (IaC)`
`Blackwell RTX 5070 + RTX 3060` `Ollama (90+ t/s) & ChromaDB`
`Hermes Swarm & SynapseFlow` `Git / Bash / Systemd`

### ▶ ENDPOINT & SERVER FLEET GOVERNANCE

`Microsoft Intune (MDM/MAM)` `Windows Autopilot (Zero-Touch)`
`SCCM / MECM Modernization` `Win32 Packaging (.intunewin)`
`58+ Proactive Remediations` `BitLocker Automated Escrow`
`Windows Server (NT4-2022)` `RHEL / Ubuntu / Debian / macOS`

### ▶ ZERO TRUST, SSO, DNSSEC & OBSERVABILITY

`Authentik Enterprise SSO/IdP` `Checkmk & Reconya L2`
`3-Node Pi-hole v6 DoH/DNSSEC` `AegisDNS (RFC 8484 DoH)`
`Nebula-Sync HA Replication` `Automated SSL (DNS-01)`
`Conditional Access / PIM` `SAML 2.0 / OIDC / FIDO2`

## // 03 ENTERPRISE INFRASTRUCTURE, CLOUD & DEVOPS EXPERIENCE

**Senior Cloud, DevOps & Infrastructure Architect** • O.S. Consulting, Inc. — Remote | 2025 – Present

- **10-Node HA Kubernetes (K3s) Orchestration**: Architected and deployed a 10-node HA K3s cluster (5 etcd control-plane quorum nodes + 5 workers, 72 vCPUs, 84GB RAM) hosting 70+ pods across 14 production namespaces (Authentik SSO, Checkmk, Pi-hole DoH/DNSSEC, Portainer, Tomcat 10.1, Reconya, PrintForge, Arr suite); engineered Cloudflare Tunnel DaemonSets, Traefik & NPM ingress, kernel sysctl optimization, and NVMe/NFS storage.
- **Sovereign AI, Dual-GPU RAG & Autonomous Agent Swarms**: Engineered a dedicated dual-GPU AI platform (NVIDIA RTX 5070 Blackwell + RTX 3060) running Ollama (90+ tok/sec Flash-Attention), Open WebUI with CUDA ChromaDB vector RAG embeddings & Faster-Whisper STT, Hermes 5-agent autonomous engineering swarm (bidirectional Telegram Bot control), and SynapseFlow visual DAG automation.
- **Proxmox VE Fleet & Workload Modernization**: Administer a 5-node Proxmox VE HA cluster, executing live LXC-to-Kubernetes migrations (Apache Tomcat 10.1, Reconya L2 discovery), configuring 12-core AVX2 host passthrough and 4GB RAM tmpfs transcode buffers, and eliminating hypervisor memory ballooning contention.
- **Tri-Node DNSSEC/DoH & Automated GitOps SSL Lifecycle**: Engineered a 3-node Pi-hole v6, Unbound recursive DNSSEC, and Cloudflare DoH sidecar cluster synced via Nebula-Sync; built automated Let's Encrypt wildcard SSL renewal pipelines via Cloudflare DNS-01 APIs and Kubernetes CronJobs.
- **Hybrid Cloud Identity & Zero-Touch Endpoint Governance**: Architect hybrid AD to Entra ID synchronizations (PHS, Seamless SSO, Conditional Access, PIM) and modernize legacy SCCM/GPO environments to Microsoft Intune with zero-touch Windows Autopilot and 58+ PowerShell proactive remediations.
- **Tier-3 Root Cause Analysis (RCA) & Observability**: Deployed Checkmk enterprise monitoring and Reconya continuous Layer-2 ARP/ICMP discovery; serve as principal escalation architect for complex cloud, hypervisor, directory, and network incidents.

**Senior Systems Administrator & Cloud Infrastructure Architect • BlueMatrix** Durham, NC | 2018 – 2025

- **Global Multi-Site Active Directory Forest (7 Data Centers)**: Architected, built, and maintained enterprise Active Directory forest and domain infrastructure across 7 global data centers and regional offices; tuned multi-site replication topology, site-cost weighting, and distributed DNS to guarantee sub-second authentication worldwide.
- **Multi-Hypervisor Virtualization & Storage Area Networks (SAN)**: Administered and tuned high-density virtualization clusters across VMware vSphere/ESXi, Linux KVM, Citrix XenServer, and XCP-NG; architected enterprise Storage Area Networks (SAN), iSCSI/Fibre Channel LUNs, shared NFS arrays, dynamic memory allocations, and automated DR snapshot replication.
- **Microsoft Azure Cloud Infrastructure & Automation**: Spearheaded phased migrations transitioning mission-critical on-premises workloads to Microsoft Azure; architected VNets, Network Security Groups (NSGs), Azure VMs, and custom Azure Automation Runbooks to govern automated compute scaling and patch compliance.
- **Automated Identity Lifecycle Engineering**: Developed enterprise PowerShell and Graph API workflows integrating HR systems with AD DS and Entra ID, automating provisioning, RBAC group delegation, M365 licensing, BitLocker escrow, and zero-day offboarding security revocation.

**DevOps Engineer & Senior Systems Administrator • StreamingEdge (Tradition America)** New York, NY (Hybrid) | 2013 – 2018

- **Financial CI/CD Release Pipeline Engineering**: Directed release engineering across a structured 5-stage deployment pipeline (DEV ➔ QA ➔ UAT ➔ Staging ➔ Production) for institutional high-frequency financial trading systems, orchestrating zero-downtime application releases, automated rollback gates, and configuration verification.
- **Mission-Critical Systems Reliability (99.999% SLA)**: Engineered and administered high-availability Windows Server (2012–2016) and Linux (RHEL, CentOS, Ubuntu) server farms hosting low-latency financial transaction engines, consistently maintaining five-nines (99.999%) SLA uptime.
- **Clustered Virtualization & Enterprise SAN Fabrics**: Managed fault-tolerant virtualization clusters across Microsoft Hyper-V and VMware vSphere backed by enterprise Storage Area Networks (SAN), configuring multi-path I/O (MPIO), multi-NIC teaming, virtual switch isolation, host failover clustering, and DR replication.
- **Enterprise Networking & Perimeter Security**: Designed and maintained high-throughput LAN/WAN routing, core switching, hardware firewalls, site-to-site IPsec VPN tunnels, and enterprise telecommunications infrastructure.

**NOC Infrastructure Engineer • Datapipe** Jersey City, NJ | 2012 – 2013

- **24×7 Managed Cloud, SAN & Hybrid Fleet Operations**: Monitored and maintained 24×7 high availability across hosted Windows Server and Unix/Linux web, application, and database server fleets backed by enterprise SAN storage within a mission-critical managed services NOC.
- **Multi-Tier Incident Triage & Database Support**: Triaged real-time infrastructure, network, and SAN fabric alerts, supported production SQL Server and MySQL environments (backups, health telemetry), and authored detailed incident post-mortems (RCAs) and operational runbooks.

**Senior Network & Systems Administrator • Digital Network Development** Peoria, IL | 1999 – 2011

- **Enterprise Windows Server & Active Directory Operations**: Commissioned, upgraded, and administered physical enterprise server farms spanning Windows NT 4.0, 2000, 2003, and 2008, managing Active Directory, Group Policy Objects, DNS, DHCP, WDS/MDT bare-metal imaging, and enterprise storage.
- **Telecommunications, Core Networking & DR**: Engineered and maintained managed Cisco switches, routers, multi-building wireless links, T1/T3 circuits, VOIP telephony platforms, and enterprise tape/NAS disaster recovery routines.

## // 04 FLAGSHIP ENGINEERING PROJECTS & OPEN SOURCE TOOLS

**NovaFlix (Creator & Architect)** github.com/digdast/novaflix
Sovereign open media platform & home theater engine (Plex/Jellyfin alternative) with 4K HDR Direct Play, HTTP 206 Range streaming, built-in remote access, and SyncPlay.

**AegisDNS (Creator & Architect)** github.com/digdast/aegisdns
Recursive DNS resolver & RFC 8484 DNS-over-HTTPS (DoH) engine traversing ICANN root hints with zero telemetry, sub-0.5ms cache, 227k+ threat sinkhole rules, and real-time UI.

**10-Node HA K3s Cluster & Zero Trust Ingress** resume.ertechlab.com
10-node K3s cluster (5-node etcd quorum + 5 workers, 72 vCPUs / 84GB RAM, 70+ pods across 14 namespaces) with Authentik SSO, Checkmk, Cloudflared DaemonSet, and HA hosting.

**Sovereign Dual-GPU AI, Hermes & SynapseFlow** github.com/digdast/synapseflow
Dual-GPU LLM & RAG platform (RTX 5070 + RTX 3060, 90+ t/s Ollama, ChromaDB) + Hermes 5-agent autonomous swarm & SynapseFlow visual DAG workflow automation engine.

**Pulsearr & Renamarr (Creator)** github.com/digdast/pulsearr
Async Python media daemon unifying Sonarr/Radarr/Prowlarr in 85MB RAM (FastAPI/SQLite WAL, 10× RAM reduction) + zero-config IMDb/TVMaze organizer with 1-click undo.

**Proxmox Suite, PrintForge 3D & Intune Catalog** github.com/digdast/Intune-scripts
5-node Proxmox VE HA optimization suite + 2-replica K3s 3D printing portal (3dprinting.ertechlab.com) + 58 open-source PowerShell Intune/Autopilot remediation modules.

## // 05 INDUSTRY CERTIFICATIONS & TECHNICAL CREDENTIALS

✓ Microsoft Certified: Azure Administrator Associate (AZ-104)
✓ Microsoft 365 Certified: Administrator Expert (MS-102)
✓ Microsoft Entra ID: Administration & Identity Security
✓ CompTIA Security+ Certified
✓ CompTIA Network+ Certified
✓ Inter-Tel University: Telecommunications & VOIP Systems